

Blue carbon site screen

Laguna de Términos, Campeche, Mexico · Prepared 2026-10-09 · Habitat model: Unet (resnet34)

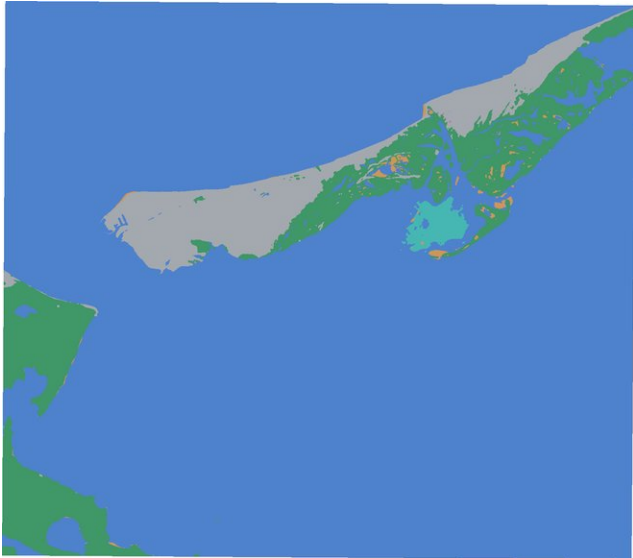
Screening verdict: **Needs investigation**

Why: Inside the Laguna de Términos Flora and Fauna Protection Area, a federal protected area decreed in 1994 (about 706,000 ha).

Satellite image and habitat map



Satellite image (Sentinel-2)



Habitat map

■ Open water ■ Mangrove ■ Salt marsh ■ Seagrass ■ Tidal flat / bare ■ Other land ■ Freshwater wetland

Habitats found

Habitat	Area (ha)	Likely range (ha)
Mangrove	7,664	7,431 – 7,896
Salt marsh	32	29 – 35
Seagrass	6,576	3,621 – 9,530

Carbon

Habitat	Stored (t CO2e)	Absorbed per year (t CO2e)	Soil carbon source
Mangrove	14,657,444 (13,808,270–15,568,456)	46,142	IPCC global default; biomass: IPCC default
Salt marsh	32,343 (30,679–34,635)	107	IPCC global default; biomass: IPCC default
Seagrass	2,689,439 (2,253,930–3,152,751)	10,690	IPCC global default; biomass: IPCC default

Credit potential (illustrative)

Basis	Low	Mid	High
Sequestration of existing habitat (credits per year)	29,059	39,202	52,535

Avoided loss, if protection stops the observed loss (credits per year)	12,276	19,547	27,682
Over 30 years (best basis)	871,775	1,176,074	1,576,056
Value per year at \$20 / \$36.5 / \$50 per credit	\$581,184	\$1,430,890	\$2,626,760

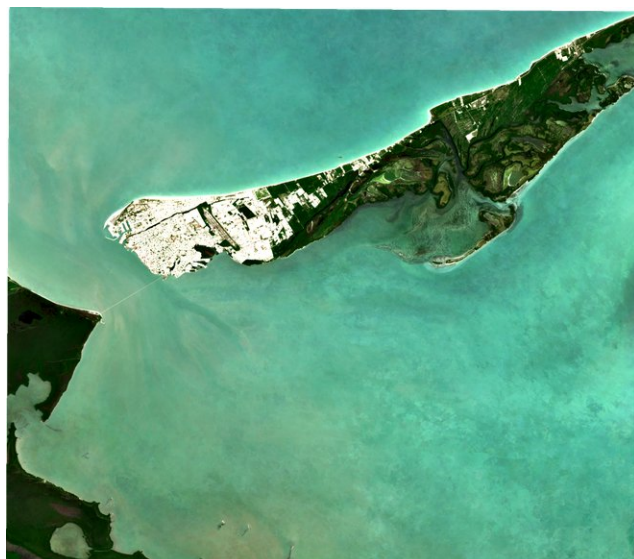
Deductions applied: risk buffer 15%, leakage 10%, uncertainty 10% (total 31%). Credits = area x carbon rate x (1 - deductions), as in the plan; real deductions come from the registry's tools and the methodology.

Sea-level rise: room to move inland

Not yet available for this site (needs elevation and built-up layers).

Change over time

Habitat	2019	2025	Change
Mangrove	7,956	7,837	-119 ha (-1%); -20 ha per year
Salt marsh	0.0	5.5	+6 ha; +1 ha per year
Seagrass	638	721	+84 ha (+13%); +14 ha per year



Rights and status screen

Check	Status	Finding
Protected area	Risk	Inside the Laguna de Términos Flora and Fauna Protection Area, a federal protected area decreed in 1994 (about 706,000 ha). Conservation credits must show the project adds protection beyond what the decree already requires (additionality). [https://en.wikipedia.org/wiki/Laguna_de_T%C3%A9rminos]
Ramsar	Caution	Ramsar Wetland of International Importance no. 1356 (designated 2 February 2004, 705,016 ha). Listed threats: sedimentation, pollution, deforestation, declining fish stocks and oil pipelines. [https://rsis Ramsar.org/ris/1356]
Mangrove law	Caution	Mexican law prohibits removing, filling or damaging mangroves (General Wildlife Law, Article 60 Ter). Legal protection already exists, so avoided-loss credits need evidence of real ongoing loss despite it. [https://www.diputados.gob.mx/LeyesBiblio/pdf/LGVS.pdf]
Tenure	Caution	Coastal land is largely federal (federal maritime-terrestrial zone) or communal (ejido). Carbon rights would come from agreements with communities and authorities, not a land purchase. Confirm with the National Agrarian Registry (RAN) and a local lawyer.
Existing projects	OK	No registered blue carbon project for Laguna de Términos was found in a web search on 2026-10-09 (one Verra-listed mangrove project exists in Nayarit). Confirm on the Verra registry before any work. [https://verra.org/projects_public_comm/aztlan-blue-carbon-mangrove-conservation-project-conserving-the-marismas-nacionales-of-nayarit]

Threats	Caution	Pemex operates oil infrastructure inside the protected area; spills and pipeline construction are recurring threats. This is evidence of pressure (relevant to additionality) and a permanence risk. [https://en.wikipedia.org/wiki/Laguna_de_T%C3%A9rminos]
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Methods and limits

- Habitats are mapped from free Sentinel-2 imagery with a deep-learning model (BlueCarbon-AI), trained on 39 coastal sites and scored on areas it never saw.
- Map accuracy on unseen areas for this habitat: F1 0.97. Areas are corrected for the model's known errors; the range shown is a plain-language margin, not a statistical interval.
- Carbon: soil carbon from measured cores where available (Smithsonian Coastal Carbon Library), otherwise IPCC 2013 Wetlands Supplement Tier 1 defaults. Mangrove biomass from NASA's canopy-height biomass map (Simard et al. 2019) where it covers the site, with IPCC carbon and root factors and a $\pm 30\%$ map error; other biomass and burial rates IPCC Tier 1.
- Seagrass is mapped conservatively and is a lower bound; murky or deep meadows are often missed.

Screening and estimates only. Not legal, financial or investment advice. Only a registry issues credits, after independent validation and verification. Confirm land tenure and carbon rights with a local lawyer.